Exercise 9 – Extending the Student Class

The school has two categories of students:

- Day students.
- Boarding students.

Both categories share the general information and functionality contained in the `Student` class. However, boarding students have additional information, such as their hostel name.

Questions

1. Create a class called `BoardingStudent` that inherits from `Student`.
2. Add an attribute called `hostelName`.
3. Create a constructor that initialises both the inherited attributes and the hostel name.
4. Add a method called `displayBoardingDetails()`.
5. Create a class called `DayStudent` that also inherits from `Student`.
6. Add an attribute called `transportMode`.
7. Create objects of both classes.
8. Demonstrate that both classes can use the methods inherited from `Student`.
9. Explain the advantages of inheritance in this example.

Challenge question: Override the `displayDetails()` method in `BoardingStudent` so that it also displays the hostel name.